

THE FOURTH AXIS

Depth, tokens, time. Three solved. The fourth is
where the money went.

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06.10.2026 SEED 7C21 K 21/4 37/5 19/6 STILL 01 OF 01

69

RECORD [08][01]

OpenAI holds 69 published patent families. Forty-nine were filed organically; the earliest priority date is mid-2022. The number on model architecture, pretraining or alignment is zero.

What the company does patent is how a model remembers between runs. A granted patent for stateful pretrained transformers. A published application for ChatGPT Memory. Granted patents for prompt caching and server-side conversation state. The brain goes on arXiv. The notebook goes to the patent office.

POSITION

Every architectural breakthrough of the last decade gave information a place to persist.

The identity path let it persist across depth. Registers let it persist across tokens. State-space memory let it persist across time. Nobody has given it a way to persist across runs. Three axes solved. The fourth is unsolved, and it is where the money went.

DEPTH

IDENTITY PATH, 2015

TOKENS

REGISTERS AND SINKS, 2023

TIME

STATE-SPACE HYBRIDS, 2025

RUNS

OPEN

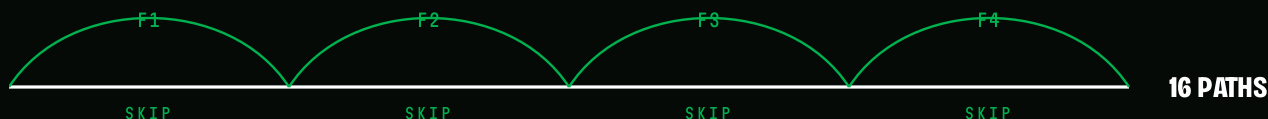
HOW TO READ

Three registers. **Measured** text carries a bracketed number pointing to a primary source on page 13. **Record** is a patent-office or filing record; numbers and dates come from the record itself, and the OpenAI numbers were compiled from Google Patents by a patent analyst, so open them before you quote them. **Position** is set condensed and bold. It is mine, stated without hedging, owned by name.

ACROSS DEPTH

MEASURED [10][16]

Plain networks stopped improving past about thirty layers. The identity path fixed it in 2015: Highway Networks in May, ResNet in December at 152 layers, a 1001-layer pre-activation ResNet on CIFAR the next year. The reason it worked was stranger than the fix. Veit and colleagues showed a residual net behaves as an ensemble of short paths. Delete a block and half the paths survive. In a 110-layer network the gradient runs through paths ten to thirty-four layers deep. The rest contribute nothing.

VEIT ET AL., 2016: A RESIDUAL NET AS 2^n PATHS [16]

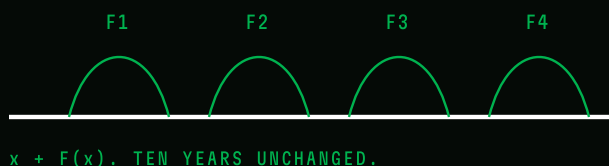
DELETE A BLOCK AND HALF THE PATHS SURVIVE. IN A 110-LAYER NET THE GRADIENT RUNS THROUGH PATHS 10 TO 34 LAYERS DEEP

MEASURED [06][07][31]

The identity path then went untouched for ten years. It is being rewired now. Hyper-connections widened the stream into n parallel streams with a learned mixing matrix [06]. DeepSeek's manifold-constrained version pins that matrix to a doubly stochastic form, which stopped the loss spikes that made the first version unstable [07]. Value residuals send first-layer values to every layer: the same loss with 13 to 16 percent fewer parameters and half the KV cache [31].

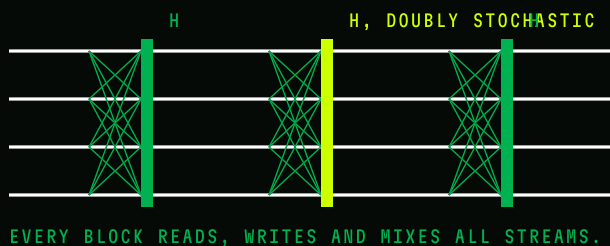
ONE STREAM

IDENTITY PATH, 2015 TO 2025



n STREAMS

HYPER-CONNECTIONS, 2024. mHC, 2026 [06][07]



SCHEMATIC. THE MIXING MATRIX IS LEARNED AND INPUT-DEPENDENT.

POSITION

The one wire nobody touched for a decade is now a switchboard. Depth is no longer a line. It is a bus.

DEPTH AS TIME

MEASURED [09]

The second rewiring treats depth as a dial. A looped transformer runs one block L times over the same tokens with the same weights. MIT's Hyperloop adds a hyper-connection per pass and matches depth-matched baselines, and the stabilized mHC baseline, with about half the parameters at 240M, 1B and 2B scale. A wave of looped and recurrent-depth papers landed between April and June 2026 on the same idea: latent reasoning, fixed-point reasoners, looped world models.



HYPERLOOP: A LOOPED MIDDLE BLOCK WITH PER-PASS HYPER-CONNECTIONS. HALF THE PARAMETERS OF A DEPTH-MATCHED BASELINE.

POSITION

If the ensemble paper was right, depth was never the thing. Paths were. A loop is a way to buy paths without buying layers.

The practical result is that a model can think longer on a hard token and shorter on an easy one. Depth, for the first time, has a knob.

50%

FEWER PARAMETERS, HYPERLOOP VS DEPTH-
MATCHED [09]

13 TO 16% 10

FEWER PARAMETERS, VALUE RESIDUALS
[31]

YEARS THE IDENTITY PATH WENT
UNCHANGED

ACROSS TOKENS

MEASURED [11]

Large vision transformers were caught repurposing their own input. About two percent of patch tokens, always over blank sky or dark corners, carried norms roughly ten times normal and had lost their own pixels. Trained as classifiers, those stolen patches named the car. The model needed somewhere to hold the whole image and took the least useful tokens it had. Four learned register tokens, discarded at output, made the artifact vanish.

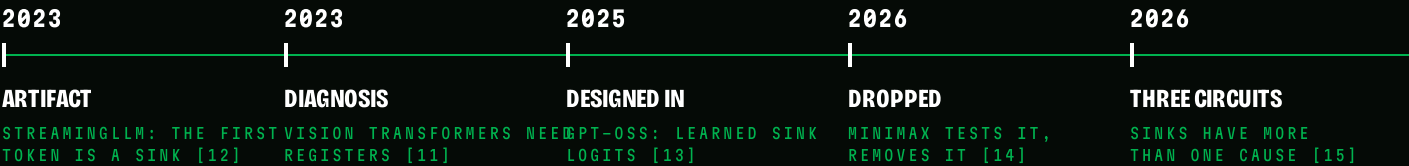
TOP-1 ON THE CARS DATASET FROM ONE TOKEN, PERCENT [11]



A BLANK-SKY PATCH THAT LOST ITS OWN PIXELS CAN NAME THE CAR. IT WAS HOLDING THE WHOLE IMAGE.

MEASURED [12] [13] [14] [15]

Language models do the same thing with the first token: attention sinks [12]. In 2025 OpenAI designed the sink in rather than leaving the model to improvise one, a learned per-head logit with no value row [13]. In 2026 MiniMax tried the same trick, found it did not suppress the first-token sink and gave no reliable gain, and removed it [14]. A mechanistic account of sinks in GPT-2 found three cooperating causes, each dispensable on its own [15].

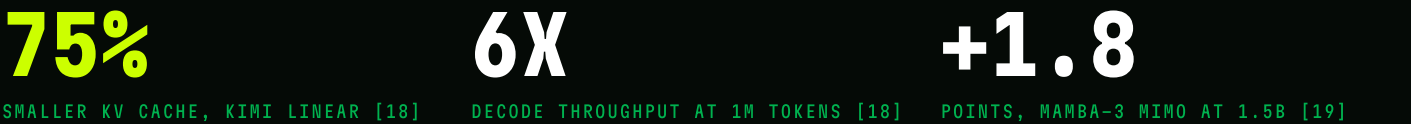


POSITION

A model with nowhere to put a thought will steal a token. Give it one on purpose. Then check that it took it.

ACROSS TIME

MEASURED [17][18][19] Persistence across time shipped as a ratio. Qwen3-Next runs 48 layers at three to one: 36 linear-attention layers carrying a delta-rule state, 12 full attention [17]. Kimi Linear cuts the KV cache by 75 percent and decodes six times faster at a million tokens [18]. Mamba-3 gives the state complex values, and parity, a state-tracking test where Mamba-2 sat near random, goes to about 100 percent; at 1.5B it adds 1.8 points of average accuracy over the next best model [19].



MEASURED [20] One lab walked it back. MiniMax-M2 runs full attention in every layer. Its technical report says no efficient-attention variant reliably matched full attention in production on reasoning, coding and agent tasks.

POSITION Time persistence is solved for the benchmark and contested for the agent. The state is cheap to carry and expensive to trust.

THEY
PATENTED
THE NOTEPAD

ACROSS RUNS

POSITION

Weights are a photograph taken the day training stopped. Everything after is read off a caption. The fourth axis is whether the photograph can change without tearing.

MEASURED
[21] [22] [23] [24] [25] [26] [27]

The attempts are serious and all partial. Titans adds a neural memory that learns at test time, gated by surprise; an independent reproduction found the memory helps but the full model does not always win [21] [22]. Google's HOPE treats its own weights as writable memory with blocks updating at different frequencies, with no public release and no forgetting curve [23]. SEAL lets a model write its own training data and apply it: on an ARC subset, 0 percent in context, 20 percent with untrained self-edits, 72.5 percent with SEAL. Repeat the edits and it forgets [24]. Retrievable Gradients store a gradient per document, apply it, then revert, so nothing drifts and nothing stays [26].

	WRITES WEIGHTS	SURVIVES THE RUN	THE CATCH
TITANS [21]	FAST WEIGHTS	WITHIN CONTEXT	REPRODUCTION CONTESTED
HOPE [23]	YES, AT INFERENCE	CLAIMED	NO RELEASE, NO CURVE
SEAL [24]	YES, BY SELF-EDIT	YES	FORGETS ON REPEAT
TTT, NEXT TOKEN [25]	FAST WEIGHTS	WITHIN CONTEXT	RESETS
RETRIEVABLE GRADIENTS [26]	APPLY, THEN REVERT	AS STORED GRADIENTS	NOT IN THE WEIGHTS
MEMORY LAYERS [27]	AT TRAINING	NO	FROZEN AFTER TRAINING

EVERY ROW IS PARTIAL. THE PARTIAL IS ALWAYS IN THE SAME COLUMN.

POSITION

Every row is partial. The partial is always in the same column. Nobody has a weight that learns on Tuesday and remembers Monday.

WHAT SHIPPED

RECORD

[01][02][03][04][05]

Persistence across runs shipped. It shipped around the weights, not in them. ChatGPT Memory is a size-capped, human-readable list of facts about the user, with reinforcement learning deciding what makes the list and background housekeeping merging duplicates [02]. Prompt caching fingerprints a prefix so the same machine reuses compute across sessions [03]. The Responses API keeps conversation state server-side and resumes from an identifier [04]. A stateful transformer pauses mid-answer, swaps in fresh instructions and resumes [01]. Groq's granted patent keeps model state resident in deterministic, compiler-scheduled memory, priority 2021, filed 2024 [05].

WEIGHTS	NOTHING SHIPPED HERE
FAST WEIGHTS	RESEARCH ONLY. TITANS, TTT, HOPE
CACHE	PROMPT CACHING PATENT [03]. NVIDIA PERSISTENT KV CACHE, PRESS [29]
NOTEPAD	CHATGPT MEMORY PATENT: A SIZE-CAPPED, HUMAN-READABLE LIST [02]
SERVER STATE	STATEFUL TRANSFORMERS [01]. RESPONSES API [04]
SILICON	GROQ DETERMINISTIC MEMORY, STATE KEPT RESIDENT [05]

WHERE PERSISTENCE ACROSS RUNS ACTUALLY LIVES, OCTOBER 2026. DEEPER IN THE STACK IS LOWER ON THE LIST.

MEASURED [28]

PRESS [29]

The software mirror: a 2026 paper formalizes session state kept across requests as a requirement for constant query latency, not an optimization [28]. NVIDIA announced a persistent, NVMe-resident KV cache as agentic long-term memory at CES 2026; press only, no patent located [29].

POSITION

Nothing shipped in the weights. Everything shipped around them. That is the market telling you where the unsolved problem is.

THE MAP

POSITION

What a lab publishes is what it is proud of. What it patents is what it sells.

PUBLISHED. NO PATENT FOUND.

TITANS	GOOGLE, 2025
MEMORY LAYERS AT SCALE	META, 2024
HYPER-CONNECTIONS	BYTEDANCE, 2024
ATTENTION SINKS	MIT AND META, 2023
REGISTERS	META, 2023
mHC	DEEPSEEK, 2026

PATENTED. NO PAPER.

STATEFUL TRANSFORMERS	OPENAI, US12400074B1
MEMORY	OPENAI, US2025200361A1
PROMPT CACHING	OPENAI, US12596764B1
RESPONSES API	OPENAI, US12591766B1
DETERMINISTIC MEMORY	GROQ, US12561279B2
CONTINUAL LEARNING	CHUNG ANG, US20260093780A1

WHAT A LAB PUBLISHES IS WHAT IT IS PROUD OF. WHAT IT PATENTS IS WHAT IT SELLS. [01] TO [05], [08]

6

ARCHITECTURES PUBLISHED, NO PATENT FOUND

6

MEMORY AND STATE PATENTS, NO PAPER

18

MONTHS FROM FILING TO PUBLICATION

RECORD [30]

The pattern holds outside the frontier labs. A Korean university filed a continual-learning method that learns only additive prompts on a frozen backbone to avoid forgetting; filed September 2025, published April 2026, no paper attached. Patent-only work is invisible to anyone who reads only arXiv.

MEASURED

No patent was located for Titans, Memory Layers, hyper-connections, attention sinks or registers. Applications publish eighteen months after filing. A Titans filing, if one exists, would not be visible yet. Absence of a record is not absence of a filing.

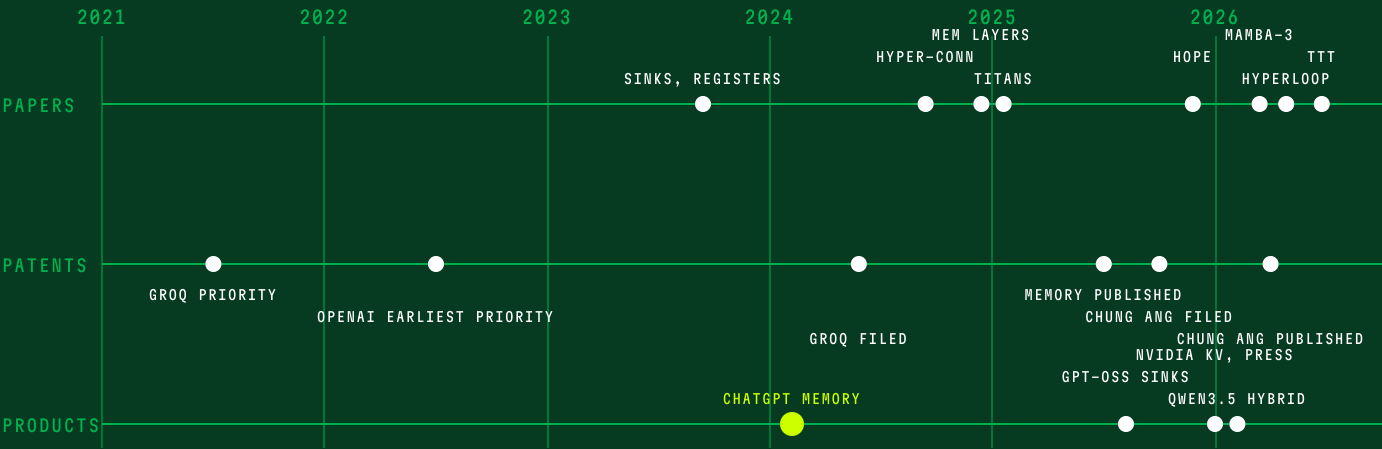
POSITION

Read the filings. They are the only place a lab tells you, under oath, what it thinks is worth owning.

THE CLOCK

RECORD
AND MEASURED

Three clocks run at different speeds. Papers post within weeks of a result. Filings surface eighteen months after priority. Products ship when the business is ready, which is sometimes before the paper exists.



A MEMORY FEATURE SHIPPED BEFORE THE FIRST TEST-TIME MEMORY PAPER WAS POSTED. THE NOTEPAD BEAT THE BRAIN TO MARKET.

2024

CHATGPT MEMORY SHIPS

2025

FIRST TEST-TIME MEMORY PAPER, TITANS [21]

2026

MEMORY FILING PUBLISHED [02]

POSITION

ChatGPT Memory shipped in 2024. The first test-time memory paper posted in January 2025. The notepad beat the brain to market by most of a year, and the gap has not closed.

If you want to know what persistence will look like in 2028, the filings of 2026 are the only forward-looking document that exists.

WHERE IT IS WRONG

A

Absence of a patent is not absence of a filing.

Eighteen months of lag means every 2025 architecture filing is still dark. The map on page 10 is a map of what is visible, and the publish-versus-patent split could narrow in 2027 as filings surface.

B

The four axes are my lens, not their taxonomy.

None of the cited papers uses the word persistence. Depth, tokens, time and runs is a synthesis. It organizes the material; it does not come from it. Treat it as a frame to argue with.

C

The fourth axis may already be moving.

HOPE writes its own weights at inference [23]. Retrievable Gradients persist across runs without drift, outside the weights [26]. One published forgetting curve that stays flat across repeated updates ends this paper's claim, and that paper could be in the NeurIPS 2026 camera-readies.

D

A filing is a hedge, not a roadmap.

Companies patent defensively. A granted memory patent proves the mechanism was worth protecting, not that it is the mechanism in production today, and a product can change under a patent without a new filing.

0

PRODUCTION MODELS LEARNING ACROSS
RUNS IN THE WEIGHTS

3

SERIOUS PARTIAL ANSWERS: HOPE, SEAL,
RETRIEVABLE GRADIENTS

1

FLAT FORGETTING CURVE WOULD END THE
ARGUMENT

SOURCES

- [01] OpenAI. Stateful pretrained transformers in a generative response engine. US12400074B1, granted. USPTO record.
- [02] OpenAI. Selective learning of information for the generation of personalized responses. US2025200361A1 / WO2025128397A1. Google Patents record.
- [03] OpenAI. Prompt caching. US12596764B1, granted. Google Patents record.
- [04] OpenAI. Responses API, server-side conversation state. US12591766B1, granted. Google Patents record.
- [05] Abts, Groq Inc. Deterministic memory for tensor streaming processors. US12561279B2, filed 3 Jun 2024, priority 7 Jul 2021. USPTO record.
- [06] Zhu et al, ByteDance. Hyper-connections. arXiv:2409.19606, ICLR 2025.
- [07] Xie et al, DeepSeek. Manifold-constrained hyper-connections (mHC). 2026.
- [08] PatentVest. What is OpenAI actually patenting? Analysis of all 69 published families. 23 Jul 2026.
- [09] Zeitoun, Torroba Hennigen, Kim, MIT. Hyperloop transformers. arXiv:2604.21254, Apr 2026.
- [10] Srivastava, Greff, Schmidhuber. Highway networks, May 2015. He et al. Deep residual learning, Dec 2015. He et al. Identity mappings in deep residual networks, 2016.
- [11] Darcet, Oquab, Mairal, Bojanowski, Meta. Vision transformers need registers. arXiv:2309.16588, ICLR 2024.
- [12] Xiao et al. Efficient streaming language models with attention sinks. arXiv:2309.17453, 2023.
- [13] OpenAI. gpt-oss model card and architecture notes, Aug 2025: learned per-head sink logits.
- [14] MiniMax. MiniMax sparse attention. arXiv:2606.13392, 2026.
- [15] A mechanistic account of attention sinks in GPT-2. arXiv:2604.14722, 2026.
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- [18] Moonshot AI. Kimi Linear. arXiv:2510.26692, 2025.
- [19] Mamba-3: improved sequence modeling using state space principles. arXiv:2603.15569, ICLR 2026.
- [20] MiniMax. MiniMax-M2 technical report. arXiv:2605.26494, 2026.
- [21] Behrouz, Zhong, Mirrokni, Google. Titans: learning to memorize at test time. arXiv:2501.00663, Jan 2025.
- [22] Titans revisited: a lightweight reimplementation and critical analysis. arXiv:2510.09551, 2025.
- [23] Behrouz et al, Google Research. Nested learning and the HOPE architecture. arXiv:2512.24695, Nov to Dec 2025.
- [24] Zweiger et al, MIT. Self-adapting language models (SEAL). arXiv:2506.10943, 2025.
- [25] Test-time training with next-token prediction. arXiv:2606.21803, 2026.
- [26] Retrievable gradients: continual post-training without cumulative weight drift. arXiv:2606.15734, 2026.
- [27] Berges et al, Meta. Memory layers at scale. arXiv:2412.09764, Dec 2024.
- [28] Attention once is all you need: efficient streaming inference with stateful transformers. arXiv:2605.13784, 2026.
- [29] NVIDIA, CES 2026: persistent NVMe-resident KV cache for agents. Trade press. Unverified; no patent located.
- [30] Kim, Jin, Chung Ang University. Continual learning method and device for multimodality data. US20260093780A1, filed 29 Sep 2025, published 2 Apr 2026. Google Patents record.
- [31] Zhou et al. Value residual learning (ResFormer, SVFormer). arXiv:2410.17897, ACL 2025.

COLOPHON

Set in Bricolage Grotesque and Martian Mono. The cover rose is computed from $r = \cos(k\theta)$, frozen at $k = 21/4, 37/5$ and $19/6$: a still from a film. Preprints dated 2026 are recent; open them before citing. Patent numbers from a patent analyst should be opened at the office record before they are repeated. Screen edition. Acid does not survive CMYK.

A decade of architecture can be read as one project with three finished parts. The identity path taught a network to carry a signal through a hundred layers without losing it. Registers taught it to hold a thought that belongs to no token. State-space memory taught it to carry that thought across a million positions.

Then the run ends, the state is discarded, and the next run starts from the photograph.

Three axes are architecture. The fourth is still a product feature. Until a weight can learn on Tuesday and keep Monday, that is the whole business.

